

Algorithm

Lecture #14

Syed Hamed Raza

Design & Analysis of Algorithms

Analysis of Quick sort average case

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- We will now show that in the average case, quicksort runs in $\Theta(n \log n)$ time.
- Recall that when we talked about average case at the beginning of the semester, we said that it depends on some assumption about the distribution of inputs.

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- However, in the case of quicksort, the analysis does not depend on the distribution of input at all.
- It only depends upon the random choices of pivots that the algorithm makes.

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- The algorithm has n random choices for the pivot element.
- Each of these choices has an equal probability $1/n$ of occurring.
- The recurrence can be written as

$$T(n) = \begin{cases} 1 & \text{if } n \leq 1 \\ \frac{1}{n} \sum_{q=1}^n (T(q-1) + T(n-q) + n) & \text{otherwise} \end{cases}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

The time $T(n)$ is the weighted sum of the times taken for various choices of q . i.e.,

$$T(n) = \left[\frac{1}{n} (T(0) + T(n-1) + n) \right. \\ \left. + \frac{1}{n} (T(1) + T(n-2) + n) \right]$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- We have not seen such a recurrence before.
- To solve it, expansion is possible but it is rather tricky.
- We will attempt a constructive induction to solve it.
- We know that we want a $\Theta(n \log n)$
- Let us assume that $T(n) \leq c n \log n$ for $n \geq 2$ where c is a Constant

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- For the base case $n = 2$ we have

$$\begin{aligned} T(n) &= \frac{1}{2} \sum_{q=1}^2 (T(q-1) + T(2-q) + 2) \\ &= \frac{1}{2} [(T(0) + T(1) + 2) + (T(1) + T(0) + 2)] \\ &= \frac{8}{2} = 4. \end{aligned}$$

Average-Case Analysis of Quick sort

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We want this to be at most $c \log n$,
i.e.,

$$T(n) \leq c \log n$$

or

$$4 \leq c \log 2$$

therefore

$$c \geq 4/(\log 2) \approx 2.88.$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- For the induction step, we assume that $n \geq 3$ and
- The induction hypothesis is that for any $n' < n$, we have $T(n') \leq cn' \log n'$
- We want to prove that it is true for $T(n)$.

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

By expanding $T(n)$ and moving the factor of n outside the sum we have

$$T(n) = \frac{1}{n} \sum_{q=1}^n (T(q-1) + T(n-q) + n)$$

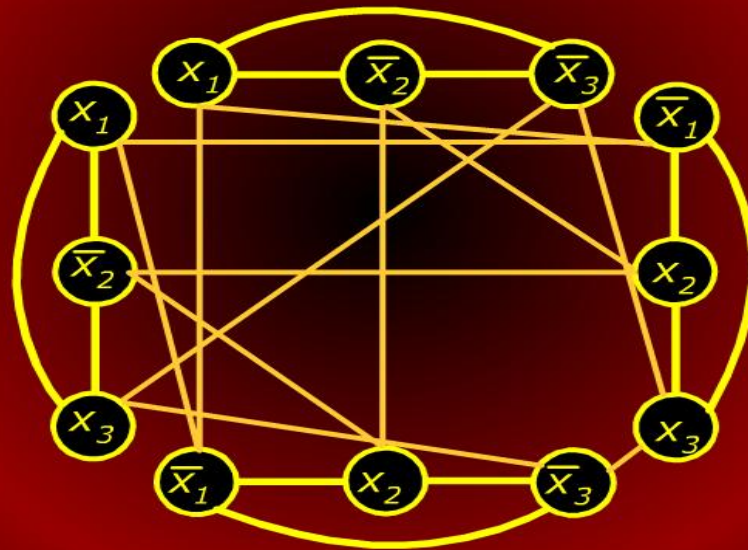
$$T(n) = \frac{1}{n} \sum_{q=1}^n (T(q-1) + T(n-q)) + n$$

$$T(n) = \frac{1}{n} \sum_{q=1}^n T(q-1) + \frac{1}{n} \sum_{q=1}^n T(n-q) + n$$

3SAT...

3SAT $\geq_P$ IS

$$(x_1 \vee \bar{x}_2 \vee \bar{x}_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_3) \wedge (x_1 \vee \bar{x}_2 \vee x_3)$$



Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

We will extract $T(0)$ and $T(1)$ and treat them specially. These two do not follow the formula.

$$\begin{aligned} T(n) &= \frac{2}{n} \left(\sum_{q=0}^{n-1} T(q) \right) + n \\ &= \frac{2}{n} \left(T(0) + T(1) + \sum_{q=2}^{n-1} T(q) \right) + n \end{aligned}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

We will apply the induction hypothesis for $q < n$ we have

$$\begin{aligned} T(n) &= \frac{2}{n} \left(T(0) + T(1) + \sum_{q=2}^{n-1} T(q) \right) + n \\ &\leq \frac{2}{n} \left(1 + 1 + \sum_{q=2}^{n-1} (cq \log q) \right) + n \\ &= \frac{2c}{n} \left(\sum_{q=2}^{n-1} (cq \log q) \right) + n + \frac{4}{n} \end{aligned}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

We have never seen this sum before:

$$S(n) = \sum_{q=2}^{n-1} (cq \ln q)$$

Recall from calculus that for any monotonically increasing function $f(x)$

$$\sum_{i=a}^{b-1} f(i) \leq \int_a^b f(x) dx$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

The function $f(x) = x \ln x$ is monotonically increasing, and so

$$S(n) = \sum_{q=2}^{n-1} (cq \ln q) \leq \int_2^n x \ln x \, dx$$

We can integrate this by parts:

$$\int_2^n x \ln x \, dx = \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) \Big|_{x=2}^n$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

$$\begin{aligned}\int_2^n x \ln x \, dx &= \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) \Big|_{x=2}^n \\ &= \left(\frac{n^2}{2} \ln n - \frac{n^2}{4} \right) - (2 \ln 2 - 1) \\ &\leq \frac{n^2}{2} \ln n - \frac{n^2}{4}\end{aligned}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

We thus have

$$S(n) = \sum_{q=2}^{n-1} (cq \log q) \leq \frac{n^2}{2} \ln n - \frac{n^2}{4}$$

Plug this back into the expression for $T(n)$ to get

$$T(n) = \frac{2c}{n} \left(\frac{n^2}{2} \ln n - \frac{n^2}{4} \right) + n + \frac{4}{n}$$

Average-Case Analysis of Quick sort

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$$T(n) = \frac{2c}{n} \left(\frac{n^2}{2} \ln n - \frac{n^2}{4} \right) + n + \frac{4}{n}$$

$$= cn \ln n - \frac{cn}{2} + n + \frac{4}{n}$$

$$= cn \ln n + n \left(1 - \frac{c}{2} \right) + \frac{4}{n}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

$$T(n) = cn \ln n + n \left(1 - \frac{c}{2}\right) + \frac{4}{n}$$

- To finish the proof, we want all of this to be at most $c n \ln n$.
- For this to happen, we will need to select c such that

$$n \left(1 - \frac{c}{2}\right) + \frac{4}{n} \leq 0$$

Average-Case Analysis of Quick sort

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If we select $c = 3$, and use the fact that $n \geq 3$ we get

$$\begin{aligned} n \left(1 - \frac{c}{2} \right) + \frac{4}{n} &= \frac{3}{n} - \frac{n}{2} \\ &\leq 1 - \frac{3}{2} = -\frac{1}{2} \leq 0 \end{aligned}$$

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

The function $f(x) = x \ln x$ is monotonically increasing, and so

$$S(n) = \sum_{q=2}^{n-1} (cq \ln q) \leq \int_2^n x \ln x \, dx$$

We can integrate this by parts:

$$\int_2^n x \ln x \, dx = \left(\frac{x^2}{2} \ln x - \frac{x^2}{4} \right) \Big|_{x=2}^n$$

Design and Analysis of Algorithms

Analysis of Quicksort

Average case

Sort Demo

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

$$\begin{aligned} T(n) = & \left[\frac{1}{n} (T(0) + T(n-1) + n) \right. \\ & + \frac{1}{n} (T(1) + T(n-2) + n) \\ & + \frac{1}{n} (T(2) + T(n-3) + n) \\ & \left. + \dots + \frac{1}{n} (T(n-1) + T(0) + n) \right] \end{aligned}$$

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- To solve it, expansion is possible but it is rather tricky.
- We will attempt a constructive induction to solve it.
- We know that we want a $\Theta(n \log n)$
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Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

- From the basis case we had $c \geq 2.88$. Choosing $c = 3$ satisfies all the constraints.
- Thus $T(n) = 3n \ln n \in \Theta(n \log n)$.

Average-Case Analysis of Quick sort

Average-case Analysis of Quicksort

$$T(n) = \frac{1}{n} \sum_{q=1}^n T(q-1) + \frac{1}{n} \sum_{q=1}^n T(n-q) + n$$

- Observe that the two sums add up the same values
 $T(0) + T(1) + \dots + T(n-1)$.
- One counts up and other counts down.
- Thus we can replace them with
 $2 \sum_{q=0}^{n-1} T(q)$.

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